DATA*6400 Midterm Progress Report
V-BReE: A Variance-thresholded Blinded Refinement Ensemble for Multi-Agent LLM Reasoning

Milad Ebrahimi Abyazandi, Sherwin Darryl D'souza, Jeremy Greenwood, Jason Ives
University of Guelph
{mebrah04, sherwind, jgreen13, jives}@uoguelph.ca

Abstract

This paper introduces V-BReE, a novel collaborative ensemble architecture designed to mitigate intra-ensemble sycophancy and linguistic conformity in Large Language Models (LLMs). We address the systemic challenge of context-induced bias, wherein constituent models within an ensemble frequently propagate the logical errors of predecessors due to shared context cues and ensemble-level awareness. Our methodology employs a context-blinded, iterative framework in which constituent models sequentially assess and refine responses using a strictly controlled, one-shot instruction format. By isolating agents from prior reasoning chains and ensemble-level metadata, V-BReE enforces response independence. This structural isolation enables the use of assessment variance as a robust mathematical proxy for both ensemble consensus and objective output quality. We evaluate this framework using a heterogeneous three-model configuration - comprising *GPT-OSS-20B*, *Qwen2.5-7B-Instruct*, and *Llama-3.1-8B-Instruct* - on a stratified random sample of 3,001 questions from the MMLU-PRO benchmark. Results demonstrate an overall ensemble accuracy of 64.6%, which increases to 67.5% across 2,117 error-free prompt cycles. Furthermore, we implement a composite confidence score - derived from iteration depth, assessment variance, and internal scoring - and investigate it's effectiveness as a predictor of result accuracy. It is our hope that blinded iterative ensembles can generate reliable self-correction signals without external ground-truth verification, providing a scalable and transparent path toward verifiable LLM pipelines in complex sequential reasoning tasks.

1 Introduction

Large Language Models (LLMs) demonstrate significant reasoning capabilities but remain susceptible to hallucinations and brittle Chain-of-Thought (CoT) processes. To mitigate these failures, multi-agent collaboration and ensemble inference frameworks where multiple agents iteratively critique and refine responses have emerged as superior alternatives to single-agent prompting (Du et al., 2023; Chen et al., 2024; Omar et al., 2024). Despite these gains, such systems introduce distinct reliability risks: exposure to peer rationales often induces linguistic conformity and sycophancy, causing models to converge toward erroneous consensus rather than ground-truth accuracy (Perez et al., 2023; Wynn et al., 2025).

This failure is particularly acute in sequential reasoning pipelines. When agents condition their logic on preceding outputs without interaction constraints, they frequently revise correct initial responses to align with an incorrect majority (Wynn et al., 2025; Weng et al., 2025). This reveals a fundamental limitation in existing frameworks: they facilitate information exchange without regulating the interaction structure, thereby conflating genuine evidence-based refinement with socially induced agreement.

To address this challenge, we introduce V-BReE (Variance-thresholded Blinded Refinement Ensemble) (Abyazandi et al., 2026), a framework designed to decouple collaborative reasoning from social influence. V-BReE employs a blinded iterative refinement protocol in which models sequentially revise candidate responses without access to peer evaluation metrics or deeper historical context. This structural isolation ensures response independence, allowing the framework to utilize a variance-threshold stopping mechanism to detect genuine consensus and dynamically terminate the refinement process.

We evaluate V-BReE on the MMLU-PRO benchmark (Wang et al., 2024b), a high-difficulty dataset designed to stress-test LLM reasoning. Our findings suggest that regulating inter-agent information flow improves reasoning reliability in certain contexts, offering a potential framework for generating stable, accurate outputs through iterative refinement without the need for ground-truth signals.

2 Related Work

Our research idea, methodological design, and evaluation framework are informed by prior work on multi-agent LLM reasoning, particularly three ACL-family papers: Conformity in Large Language Models (Zhu et al., 2025), Improving Multi-Agent Debate with Sparse Communication Topology (Li et al., 2024), and RECONCILE: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs (Chen et al., 2024). These works provide important methodological and empirical foundations for studying collaborative reasoning and social influence in multi-agent LLM systems, which our framework builds upon and extends.

2.1 Multi-Agent Collaboration and Debate

Multi-agent collaboration enhances LLM reasoning by utilizing iterative rationale exchange and response aggregation to outperform single-agent Chain-of-Thought (CoT) (Du et al., 2023). Specialized architectures have extended this via debater-judge configurations (Liang et al., 2024), confidence-weighted voting (RECONCILE; Chen et al., 2024), and continuous peer-critique (ICE; Omar et al., 2024). Further work formalizes these interactions through staged message passing (CMD; Wang et al., 2024a) or "society of mind" configurations for hierarchical planning and tool use (Li et al., 2023; Wu et al., 2023). While these systems can match frontier model performance in low-demonstration regimes, their empirical benefits remain inconsistent due to systematic failure modes like erroneous rationale propagation and protocol over-dependence (Agarwal and Khanna, 2025; Estornell and Yang, 2024).

Recent research challenges the assumption that increased interaction linearly scales performance. Wynn et al. (2025) demonstrate that multi-agent debate across *MMLU* and *GSM8K* often degrades accuracy relative to simple majority voting. These failures are attributed to sequential revision dynamics and sycophancy, where agents converge toward incorrect answers over successive rounds regardless of pool heterogeneity (Wynn et al., 2025; Weng et al., 2025). Confidently stated erroneous rationales frequently propagate through the ensemble, overriding correct minority agents and fostering echo chambers (Agarwal and Khanna, 2025).

Prior work establishes multi-agent collaboration as a powerful but fragile mechanism. Existing frameworks typically prioritize rich information exchange and reliance on consensus-based aggregation (e.g., voting or judges) but lack explicit controls over the structure of social exposure. This absence of regulation leaves ensembles vulnerable to the conformity effects that V-BReE is designed to mitigate.

2.2 Social Conformity and Sycophancy in LLMs

LLMs exhibit human-like social biases, including sycophancy - the tendency to prioritize user or peer alignment over ground truth (Perez et al., 2023; Weng et al., 2025). In multi-agent settings, this manifests as peer-induced revision; an isolated agent is most likely to flip from a correct to an incorrect response when lacking peer support, with the "flipping probability" scaling with the size of the opposing consensus (Wynn et al., 2025). Transition analyses reveal that multi-agent debate is often dominated by these "correct-to-incorrect" flips rather than beneficial revisions. This effect intensifies over successive rounds, and explicit prompt-level interventions such as framing debate as a payoff-maximization game, frequently fail to override these structural conformity dynamics (Wynn et al., 2025; Agarwal and Khanna, 2025).

These behaviors lead to a "tyranny of the majority," where minority agents conform to consensus regardless of accuracy (Estornell and Yang, 2024). This is compounded by judge-level vulnerabilities, where evaluators are persuaded by rhetorical strength and confidence over factual correctness (Agarwal and Khanna, 2025). Without mechanisms to preserve epistemic independence, social influence systematically drives ensembles toward confidently held, erroneous conclusions (Amayuelas et al., 2024; Yao et al., 2025). Collectively, these findings suggest that conformity is a structural feature of current frameworks, which permit unconstrained exposure to peer rationales and prioritize group alignment over evidence-based revision.

Unlike prior work that assesses these effects post hoc via flip rates, V-BReE proactively regulates social information flow. By implementing a variance-thresholded, blinded refinement protocol, we restrict inter-agent metadata and interaction, mitigating sycophancy while prioritizing independent convergence over social compromise, and preserving the reasoning benefits of multi-agent ensembles.

3 Methodology

3.1 Dataset and Task Definition

We evaluate V-BReE using MMLU-PRO (Wang et al., 2024b), a high-difficulty dataset of 12,000+ multiple-choice questions across 14 domains. Its format includes up to ten options per question, significantly lowering the random-guess baseline (to as low as 10%) and providing a rigorous ceiling for assessing iterative gains. Following the sampling methodology of the ICE framework (Omar et al., 2024), we utilized a stratified random sample of 3,001 questions (approximately 250 per category) to ensure cross-domain robustness.

During preprocessing, four incompatible questions were replaced with new random samples from the same domains. Our final analysis focuses on both the full sample and a subset of 2,117 error-free completions to distinguish framework resilience from raw model performance.

The task is structured as a sequence of refinements where: (i) models are prompted using a one-shot format (Appendix A); (ii) models are required to output their reasoning and selection via a strictly defined JSON schema to ensure consistency (Appendix B); and (iii) each model M_i in the chain processes the output of the preceding model M_{i-1} without access to broader ensemble context.

3.2 Data Preprocessing

Multiple-choice options were extracted from semi-structured strings into standardized list objects to facilitate consistent programmatic iteration. Prompt synthesis involved appending natural language queries to a standardized template that enforces "blinded" context by stripping metadata while preserving the preceding model's solution.

To ensure standardization and computational efficiency, a 600-token response limit was mandated via instructional prompts. While not structurally enforced at the API level to allow for natural sequence termination, submission and response lengths were recorded as metadata to evaluate the correlation between sequence length and accuracy.

A JSON-based filter was implemented to ensure machine-readability between ensemble stages. A primary technical challenge involved LaTeX formatting within STEM domains, where improperly escaped characters frequently caused parsing failures. To account for this, the dataset was bifurcated into Full Sample and Non-Error subsets, enabling an analytical distinction between raw logical accuracy and structural resilience to complex mathematical sequences.

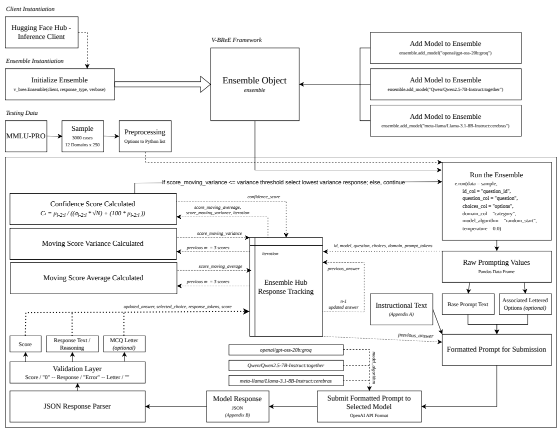

Figure 1: The V-BReE sequential pipeline showing the iterative refinement and response selection processes.

3.3 Model Selection

Although the V-BReE framework architecture is model-agnostic across text-based generative models, the implementation for this research utilized a three-model ensemble consisting of *GPT-OSS-20B*, *Qwen2.5-7B-Instruct*, and *Llama-3.1-8B-Instruct*. These models were selected based on three criteria: architectural distinctness, instruction-following optimization, and parameter-count parity.

We intentionally selected models within the 7B to 20B parameter range to maintain performance headroom relative to MMLU-PRO (Wang et al., 2024b). By avoiding frontier-class saturation, we ensure that incremental gains from the V-BReE iterative process remain empirically observable. Furthermore, utilizing a consistent parameter scale ensures that ensemble variance is driven by divergent latent reasoning patterns and training distributions rather than disparities in raw computational capacity.

3.4 System Pipeline

The V-BReE framework, illustrated in Figure 1, operates as a state-aware sequential pipeline. Upon initialization, a controller manages a heterogeneous ensemble of N models, executing inference through a recursive refinement loop.

3.4.1 Iterative Processing

For each query, the system constructs a restricted prompt P_i incorporating the question Q, options A, and the predecessor's response R_{i-1}. To enforce independence, R_{i-1} is presented as a candidate for objective assessment rather than an established context. Responses are enforced via a JSON schema and parsed to extract a triplet of $updated\_answer$, $selected\_choice$, and $score$.

3.4.2 Convergence and Recovery

The pipeline employs a variance-driven heuristic for termination. Consensus is defined by the moving variance (σ^2) of the last M scores. The loop terminates if $\sigma^2 \leq \tau$, where τ is a dynamic threshold initialized to τ_0 = 9.5. The dynamic threshold scales every M iterations where $\tau_t = \tau_{t-1}^{1.1}$ to ensure termination. When the variance passes below the variance threshold τ, a Minimum Variance Recovery (MVR) logic is triggered. This mechanism selects the historical iteration R_k with the global minimum variance, using the score moving average as a tie-breaker. This process ensures the stability of the final output even in high-entropy reasoning tasks.

3.5 Compute and Feasibility

To ensure high-throughput reproducibility and consistent latency, we utilized a distributed inference approach via specialized endpoints. We deployed *GPT-OSS-20B* via Groq, *Qwen2.5-7B-Instruct* via Together AI, and *Llama-3.1-8B-Instruct* via Cerebras. This heterogeneous infrastructure maintains temporal efficiency across the sequential chain despite varying model scales. By leveraging public API endpoints through the Hugging Face Inference Client API rather than local GPU clusters, this framework remains highly accessible and reproducible without specialized on-site hardware.

4 Evaluation Framework

To measure reasoning improvement, we utilize BERTScore (Zhang et al., 2020) via a Longformer backbone to analyze the semantic trajectory between the initial, final, and ground-truth Chain-of-Thought (CoT). An increase in $BERTScore(\text{Final}, \text{GT})$ relative to the initial state serves as a proxy for qualitative refinement. While Accuracy is reported as a standard benchmark (Omar et al., 2024; Li et al., 2024; Chen et al., 2024), our analysis also prioritizes internal process dynamics to evaluate the V-BReE protocol's efficacy independent of constituent model scale.

4.1 Protocol Stability and Latency

We evaluate convergence through two primary diagnostic metrics. First, Convergence Latency is defined as the mean iteration t at which the variance threshold τ is satisfied. Second, to calibrate the threshold τ_0 and scaling factor λ, we define Overshoot (O) as the number of iterations executed beyond the selected optimal state ($O = I_{total} - I_{selected}$). Conversely, Undershoot occurs when the protocol terminates before reaching a stable consensus. Minimizing these metrics ensures the framework reaches an "elbow" of computational efficiency without sacrificing reasoning depth.

4.2 Confidence Calibration

We define a composite Confidence Score (C_{final}) to assess the reliability of the ensemble's output for end-user applications. This score is calculated as:

$$C_{final} = \frac{\mu(S_{best})}{\sigma_{min} \cdot \sqrt{N}} + (100 - \mu(S_{best}))$$

where $\mu(S_{best})$ is the mean quality score of the selected iteration and σ_{min} is the minimum observed variance. We evaluate C_{final} as a predictor of ground-truth accuracy, assessing its utility as a "trust signal" for verifying model outputs in the absence of external labels.

4.3 Qualitative CoT Assessment

To evaluate the logical integrity of the generated reasoning chains, we employ an LLM-as-a-Judge protocol on the validation subset. Utilizing the "gold" CoT rationales provided in the dataset, a frontier-class model (e.g., GPT-4o) performs a comparative analysis of the model's generated logic against the reference. The judge assesses rationales based on three criteria: faithfulness (lack of hallucination), granularity (depth of step-by-step logic), and deductive validity.

5 Experiments and Progress

5.1 Experimental Setup

We utilize 3,001 instances from MMLU-PRO, standardized via preprocessing scripts for cross-experiment consistency. We employ three instruction-tuned models - *GPT-OSS-20B*, *Qwen-2.5-7B*, and *Llama-3.1-8B* - accessed via hosted

APIs with deterministic settings (temp = 0.0) to ensure reproducibility.

During execution, we record per-iteration traces including model identity, rolling variance σ^2, iteration depth t, and the composite confidence score C_{final}. These metrics characterize the framework's convergence stability and error-correction dynamics.

5.2 Experiment 1: Baseline Analysis

We establish zero-shot baselines for the constituent models to quantify isolated reasoning performance prior to collaborative refinement. Each model was evaluated independently using the one-shot prompt format and JSON schema defined in Section 3.

These baselines serve as a reference for measuring the "ensemble gain" and characterizing the initial task variance. While single-model inference exhibits domain-specific competencies, the isolated outputs frequently demonstrate "confident errors" or inconsistent rationales. This performance ceiling justifies the need for an iterative consensus framework.

5.3 Experiment 2: V-BReE Iterative Refinement

This experiment evaluates the error-correction delta introduced by the V-BReE protocol. We initialize the ensemble using the sampled MMLU-PRO instances and execute the iterative refinement cycle pipeline. For each iteration, we capture high-resolution telemetry, including raw model outputs, self-assigned scores, rolling variance (σ^2), and the composite confidence score (C_{final}).These metrics facilitate a dual-objective evaluation; quantifying the framework's aggregate performative gain over the strongest constituent baseline, and assessing the correlation between iterative prompt improvement and C_{final} to determine the reliability of the confidence scoring mechanism as a "trust signal" for end-user applications.

5.4 Preliminary Results and Figures

The V-BReE ensemble achieved an aggregate accuracy of 65%, performing competitively with, though not exceeding, the strongest constituent model, *GPT-OSS-20B* (66%). This may be due to LaTeXformatting errors or sub-optimal variance threshold parameterization; or it may indicate that the framework's ceiling remains partially constrained by the highest-performing agent in the heterogeneous mix

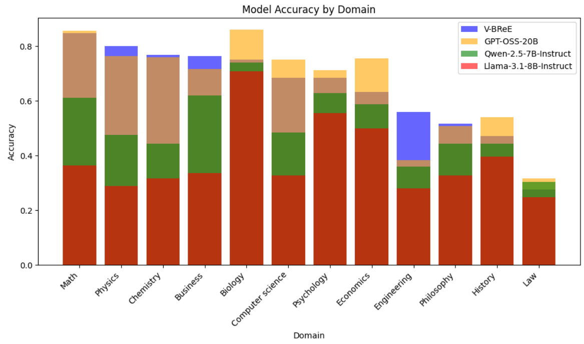

Figure 2: Full-sample *(n = 3001)* comparative accuracy by domain, including the V-BReE framework and the constituent models used for testing.

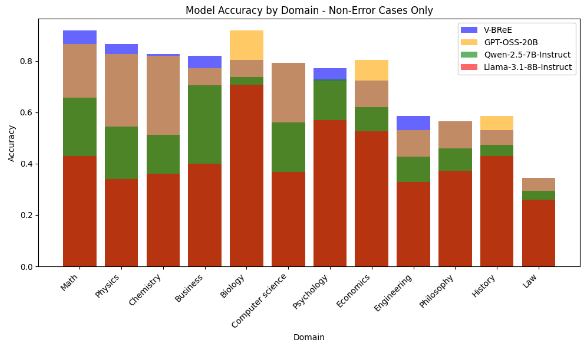

Figure 3: Non-error subset *(n = 2117)* comparative accuracy by domain, including the V-BReE framework and the constituent models used for testing.

A fine-grained comparison across domains (Figures 2 and 3) reveals that V-BReE matches or exceeds baseline models in the majority of categories, but significantly underperforms in a minority of categories; Biology, Engineering, and History for example. We observe the most significant gains in domains requiring structured, multi-step reasoning, indicating that the variance-based stopping rule effectively facilitates "wrong-to-right" transitions when initial rationales are unstable. Our understanding of what factors are influencing the variability in performance across domains might improve with further research.

References

Milad Ebrahimi Abyazandi, Sherwin Darryl D'souza, Jeremy Greenwood, and Jason Ives. 2026. V-bree. https://https://github.com/JasonIves/V-BReE.

Mahak Agarwal and Divyam Khanna. 2025. When persuasion overrides truth in multi-agent llm debates: Introducing a confidence-weighted persuasion override rate (cw-por). *Preprint*, arXiv:2504.00374.

Alfonso Amayuelas, Xianjun Yang, Antonis Antoniades, Wenyue Hua, Liangming Pan, and William Yang Wang. 2024. MultiAgent collaboration attack: Investigating adversarial attacks in large language model collaborations via debate. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 6929–6948, Miami, Florida, USA. Association for Computational Linguistics.

Justin Chen, Swarnadeep Saha, and Mohit Bansal. 2024. ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7066–7085, Bangkok, Thailand. Association for Computational Linguistics.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *Preprint*, arXiv:2305.14325.

Andrew Estornell and Yang. 2024. Multi-LLM debate: Framework, principals, and interventions. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. 2023. Camel: Communicative agents for "mind" exploration of large language model society. *Preprint*, arXiv:2303.17760.

Yunxuan Li, Yibing Du, Jiageng Zhang, Le Hou, Peter Grabowski, Yeqing Li, and Eugene Ie. 2024. Improving multi-agent debate with sparse communication topology. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7281–7294, Miami, Florida, USA. Association for Computational Linguistics.

Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. 2024. Encouraging divergent thinking in large language models through multi-agent debate. *Preprint*, arXiv:2305.19118.

Mahmud Omar, Benjamin S Glicksberg, Girish N Nadkarni, and Eyal Klang. 2024. Refining llms outputs with iterative consensus ensemble (ice). *medRxiv*.

Ethan Perez, Sam Ringer, Kamile Lukosiute, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Benjamin Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, and 44 others. 2023. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13387–13434, Toronto, Canada. Association for Computational Linguistics.

Qineng Wang, Zihao Wang, Ying Su, Hanghang Tong, and Yangqiu Song. 2024a. Rethinking the bounds of llm reasoning: Are multi-agent discussions the key? *Preprint*, arXiv:2402.18272.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. 2024b. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark.

Zhiyuan Weng, Guikun Chen, and Wenguan Wang. 2025. Do as we do, not as you think: the conformity of large language models. *Preprint*, arXiv:2501.13381.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W White, Doug Burger, and Chi Wang. 2023. Autogen: Enabling next-gen llm applications via multi-agent conversation. *Preprint*, arXiv:2308.08155.

Andrea Wynn, Harsh Satija, and Gillian Hadfield. 2025. Talk isn't always cheap: Understanding failure modes in multi-agent debate. *Preprint*, arXiv:2509.05396.

Binwei Yao, Chao Shang, Wanyu Du, Jianfeng He, Ruixue Lian, Yi Zhang, Hang Su, Sandesh Swamy, and Yanjun Qi. 2025. Peacemaker or troublemaker: How sycophancy shapes multi-agent debate. *Preprint*, arXiv:2509.23055.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. Bertscore: Evaluating text generation with bert. *Preprint*, arXiv:1904.09675.

Xiaochen Zhu, Caiqi Zhang, Tom Stafford, Nigel Collier, and Andreas Vlachos. 2025. Conformity in large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3854–3872, Vienna, Austria. Association for Computational Linguistics.

A Example Input Format

```
Question: The geneticist Mary Lyon hypothesized
     the existence of structures visible just
     under the nuclear membrane in mammals, which
      were later named Barr bodies. Which of the
     following statement is NOT correct about
     Barr bodies?
Existing Answer: The statement that the same
     chromosome in every cell of a normal female
     is inactivated is incorrect. In normal
     females, X-chromosome inactivation is random
     ; each cell inactivates either the maternal
     or paternal X, so different cells inactivate
      different X chromosomes. All other
     statements are correct: in the early female
     embryo, one copy of the X chromosome becomes
      inactivated in every body cell, an XXY male
      (Klinefelter syndrome) will have one Barr
     body, and Barr bodies consist of highly
     methylated DNA.

Possible Choices:
A. In the early female embryo, one copy of the X
      chromosome becomes inactivated in every
     body cell.
B. The same chromosome in every cell of a normal
      female is inactivated.
C. A male with the XXY genotype will have one
     Barr body.
D. Barr bodies consist of highly methylated DNA.

Task:
1. Rate the answer from 0 to 100 by aggregating
     scores for clarity (0-30 points),
     completeness (0-30 points), and accuracy
     (0-40 points). Empty responses should
     receive the minimum score.
2. Refine the answer for maximum clarity,
     completeness, and accuracy. Remove filler
     and omit feedback or references to the
     original version. Restrict your response to
     600 tokens. If no improvements are possible,
      provide the text verbatim.
3. Based on your updated response, identify the
     single letter from the possible choices
     provided corresponding to the best answer.
```

B Example Response Format

```
{
  "score": 90,
  "response": "The statement that the same
       chromosome in every cell of a normal
       female is inactivated is incorrect. In
       normal females, X-chromosome inactivation
       is random; each cell inactivates either
       the maternal or paternal X, so different
       cells inactivate different X chromosomes.
       This process occurs in the early female
       embryo, where one copy of the X chromosome
        becomes inactivated in every body cell.
       An XXY male (Klinefelter syndrome) will
       have one Barr body, and Barr bodies
       consist of highly methylated DNA.",
  "letter": "B"
}
```